

ADVANCED COGNITIVE NEUROSCIENCE

MULTIVARIATE ANALYSIS

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PORTFOLIO: VIDEO EXPLAINER

- Deadline Tuesday 23:59



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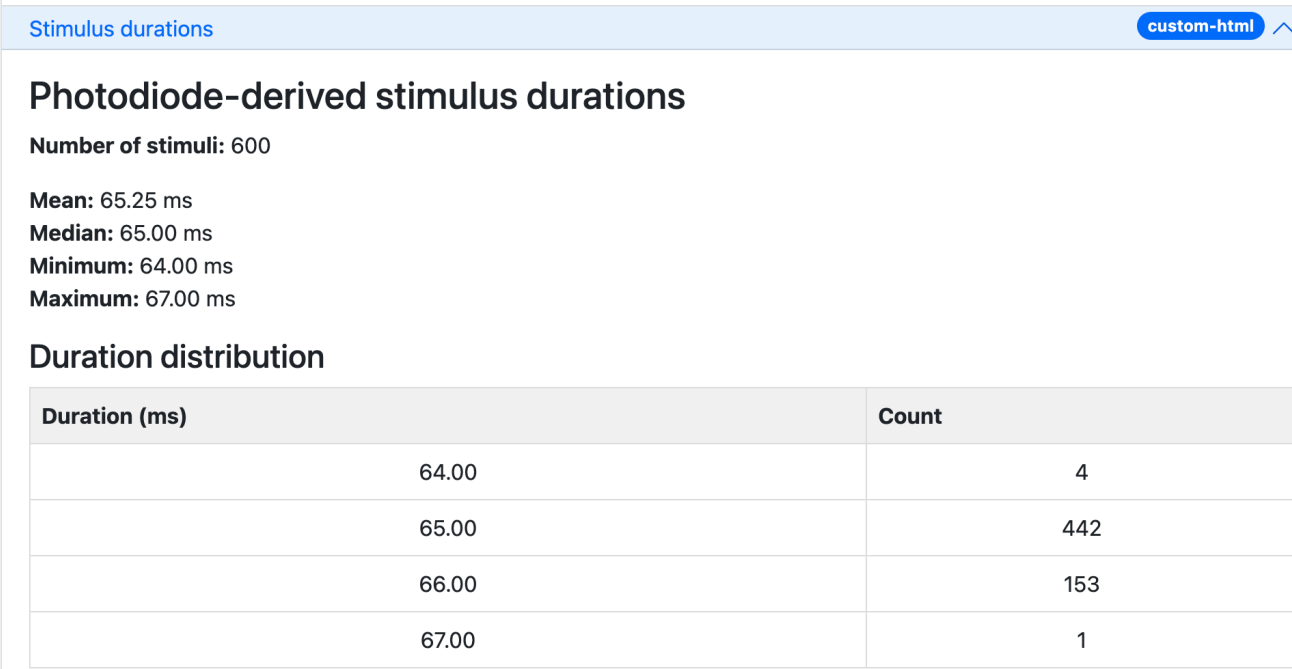
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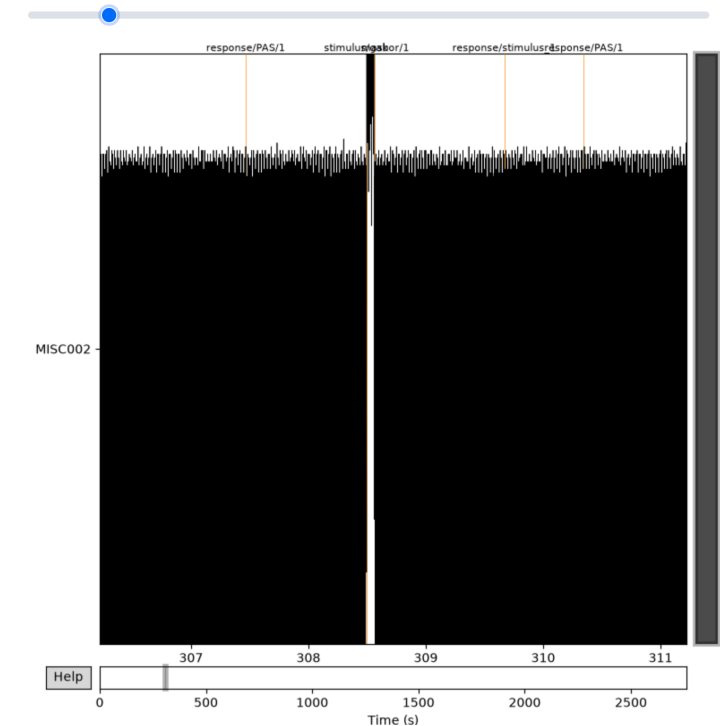


INFO

- Stimulus duration not ~33 ms as expected
- Projector likely running at 60 Hz instead of 120 Hz
 - 4 frames at 60 Hz vs at 120 Hz



```
utils/photodiode.py  
  
falling_edges() <- thanks  
to study group 1 !!!
```



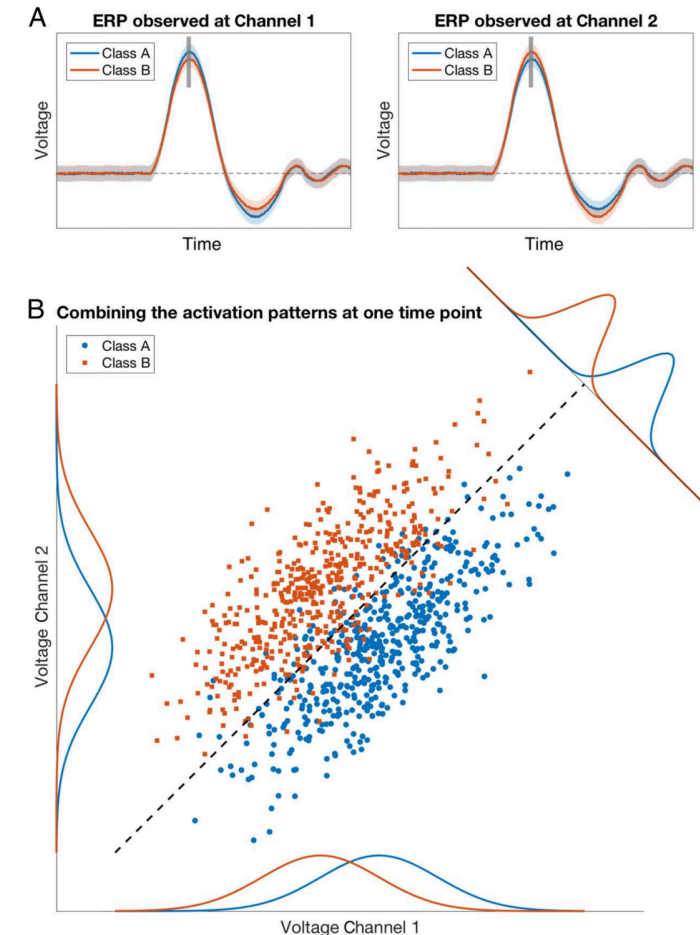
WHY USE CLASSIFICATION?

- Different questions
 - Univariate vs multivariate
 - Directional vs non-directional
 - Activation vs information
- Highly sensitive and powerful
- Spatially unbiased & data driven



UNIVARIATE VS MULTIVARIATE

- Single variable
 - No significant difference between **A** & **B** in either channel
- Multiple variables
 - Information in the two channels combined about **A** & **B**
 - Increase sensitivity

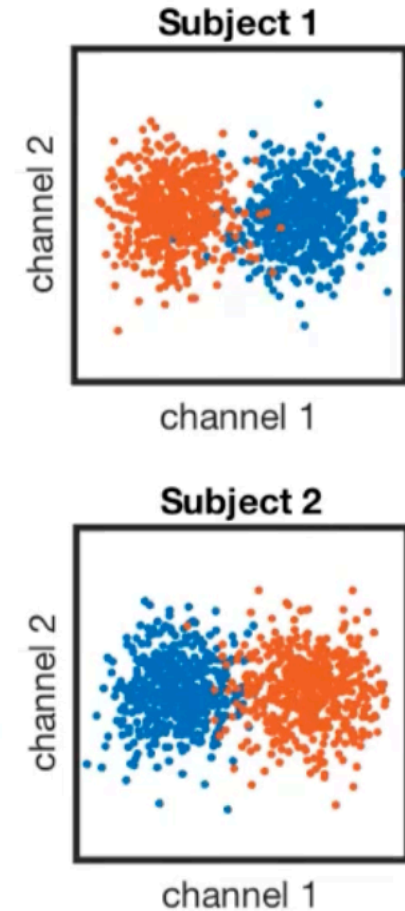


Tijl Grootswagers et al (2017). Decoding Dynamic Brain Patterns from Evoked Responses: A Tutorial on Multivariate Pattern Analysis Applied to Time Series Neuroimaging Data.



DIRECTIONAL VS NON-DIRECTIONAL

- Univariate analysis is directional
 - Subject 1 & channel 1 $A < B$
 - Subject 2 & channel 1 $A > B$
 - **No group level effect**
- Multivariate analysis analysis is non-directional
 - For both subjects we would find that $A \neq B$
 - **Group level effect**



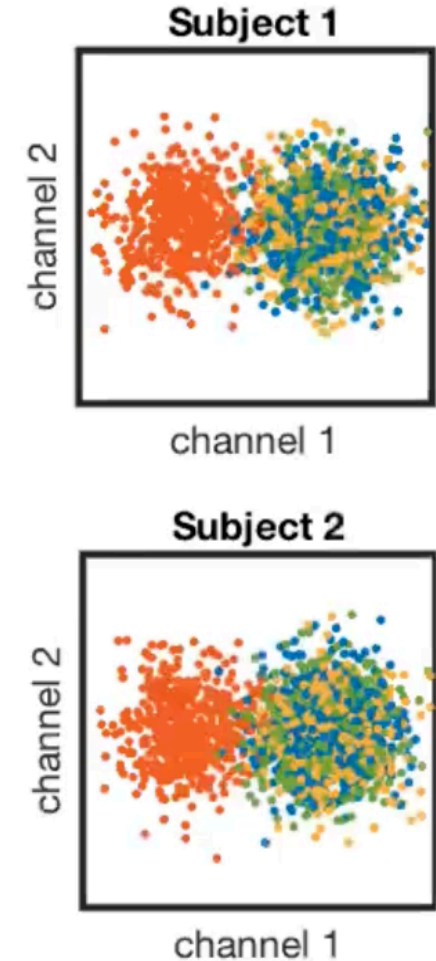
<https://www.youtube.com/watch?v=Po2XU3nMG-M>

(4 min 48 sec)



ACTIVATION VS INFORMATION

- Activation: Increased brain activity for **B**, **C** & **D**
 - Response reflects **B**, **C** & **D**
- Information: **A** is different from **B**, **C** & **D**
 - Response encodes information about **A**



<https://www.youtube.com/watch?v=Po2XU3nMG-M>

(5 min 26 sec)



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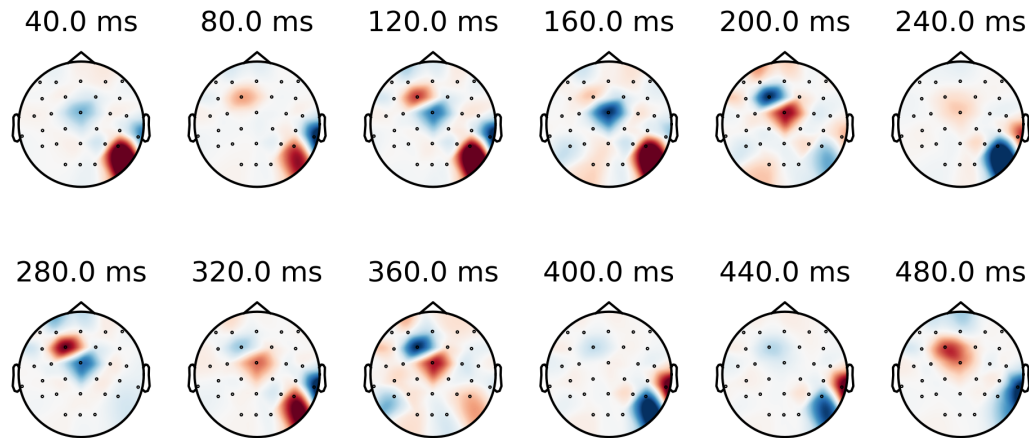
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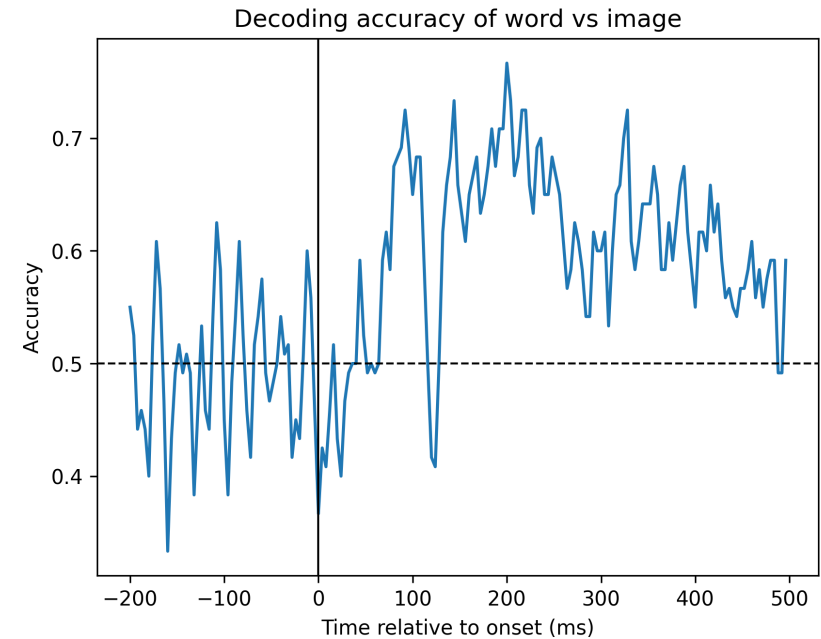


SPATIALLY UNBIASED AND DATA-DRIVEN

- No need for ROI selection, but rather letting the classifier find information
- Plotting of classifier weights (e.g. how important is data from this given sensor for classification)



https://github.com/laurabpaulsen/CogNeuro2025/blob/main/EEG_LAB/07-decoding.ipynb



DECODING FOR PREDICTION OR INTERPRETATION

Hebart & Baker (2017)

Prediction

- Goal: predict variable from MEG activity
- Any information that improves prediction is useful
- “Confounds” can still be useful for prediction
- Focus: generalisation to new data

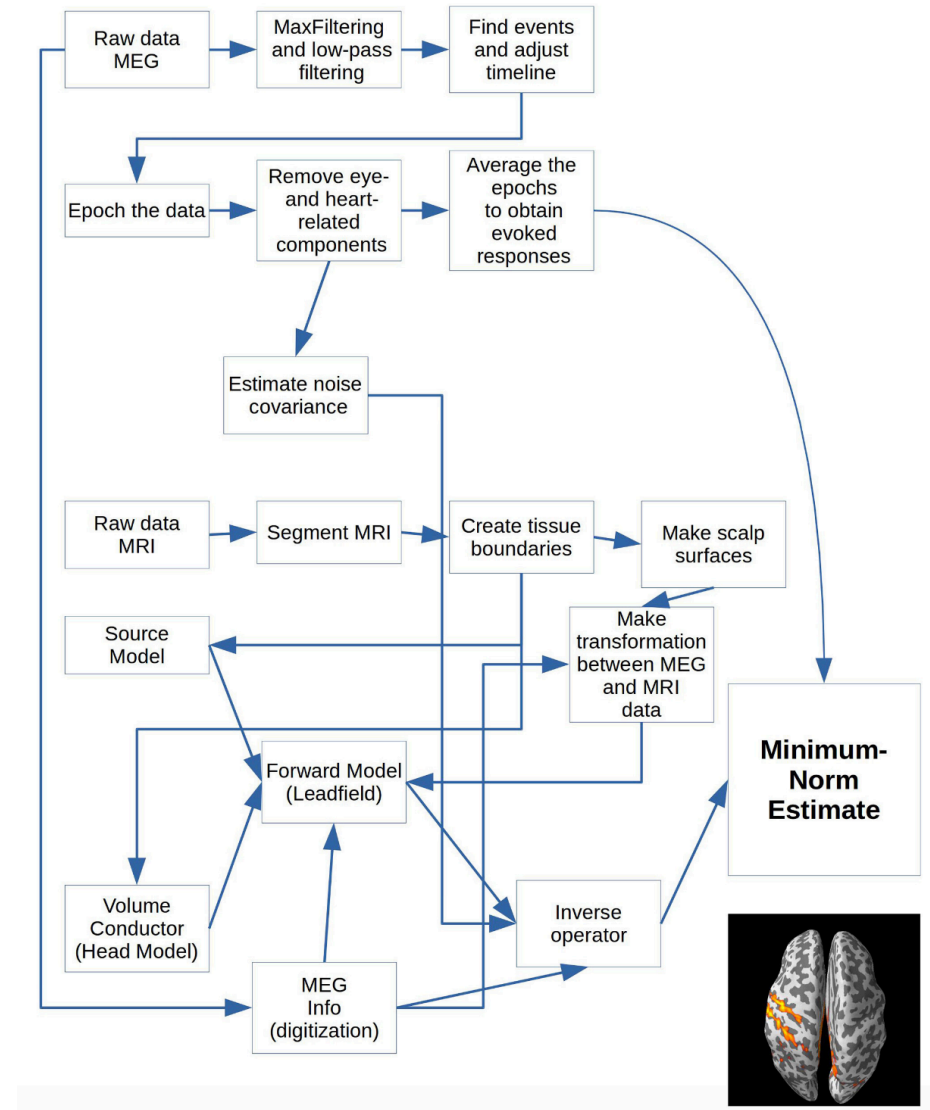
Interpretation

- Goal: understand brain function
- Need to establish where the information is coming from
- Confounds can lead to misleading neural interpretations
- Focus: what the decoding tells us about the brain



BASIC PIPELINE

- Preprocessing -> covered in previous coding classes and lectures
- Decoding
 - Classifier
 - Cross-validation
 - Inference



TRAINING A CLASSIFIER

- Input: MEG data and labels
- Extra preparation
 - Centering and scaling for each channel separately
 - Regressing out effect of “confounds”
 - More on this on Wednesday!
- Fitting the classifier

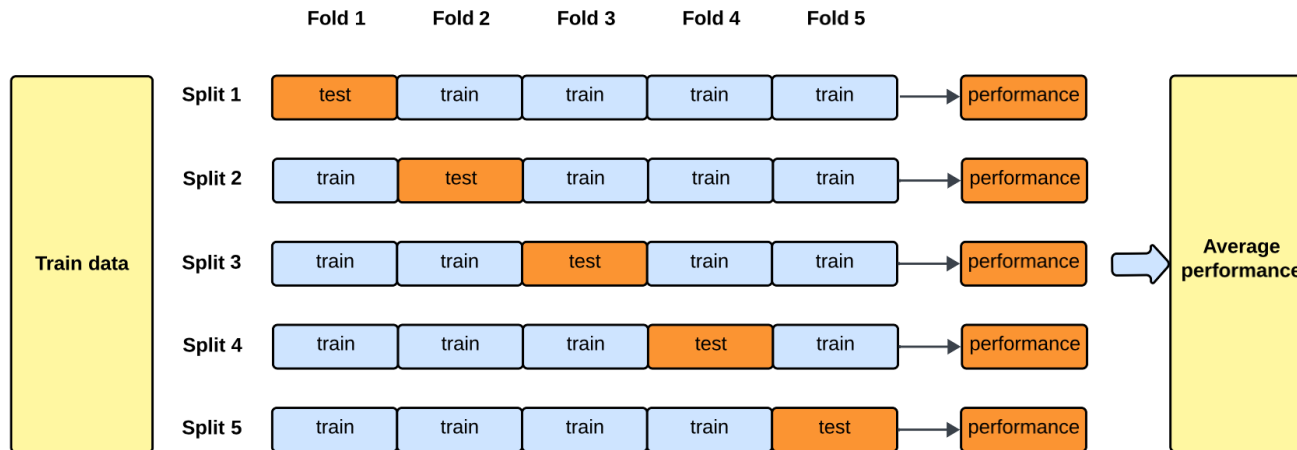
```
scaler = StandardScaler()  
X_scaled = scaler.fit_transform(X)
```

```
clf = LogisticRegression(  
    random_state=99,  
    C=1.0,  
    max_iter=1000  
)  
  
clf.fit(X, y)
```



CROSS-VALIDATION

- To avoid overfitting and enable prediction due to noise we want to test on unseen data
- Repeat training and testing on different splits of the data
- Calculate the mean cross-validated classification accuracy



<https://towardsdatascience.com/how-to-cross-validation-with-time-series-data-9802a06272c6/>

```
kfold = StratifiedKFold(5, shuffle=True)
```

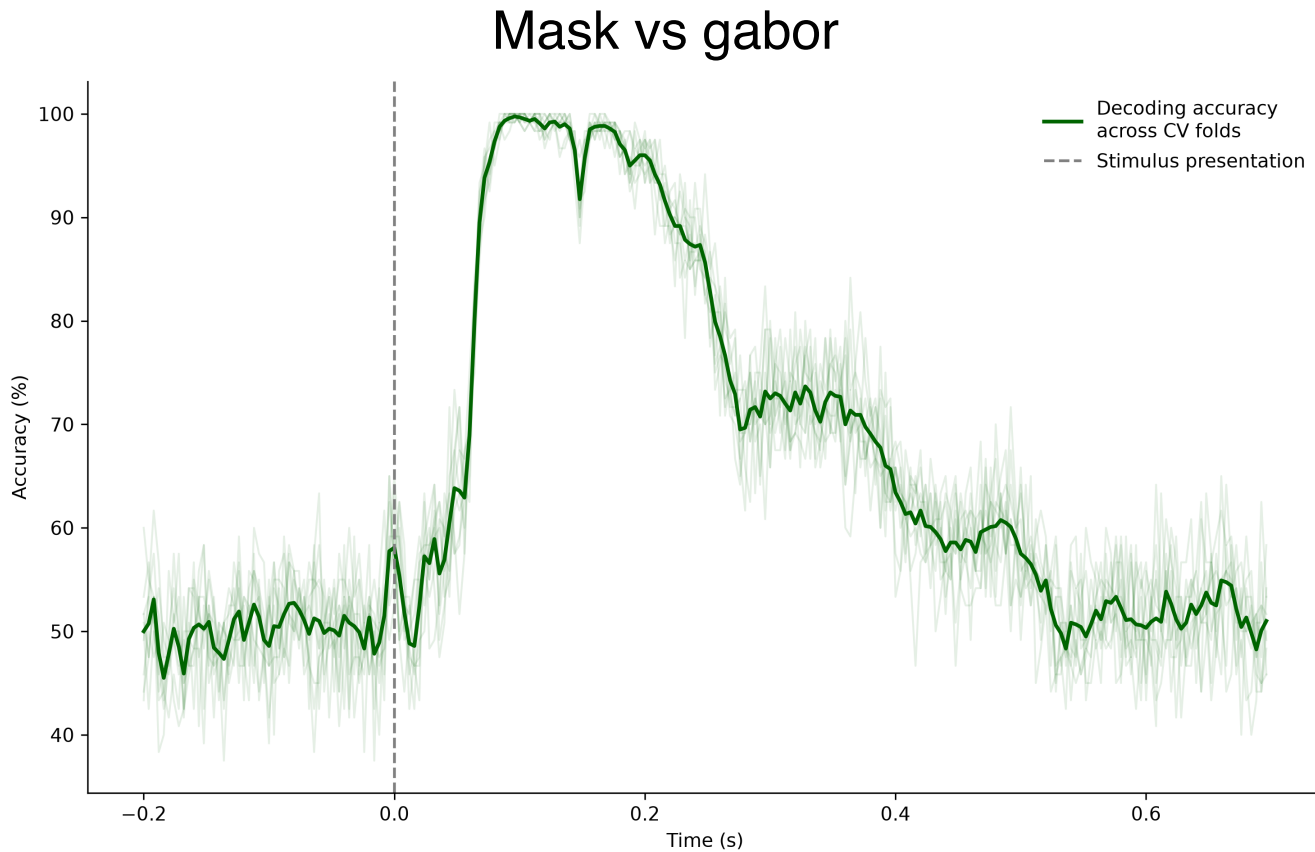
```
pipeline = Pipeline(  
    [("scaler", scaler), ("clf", clf)]  
)
```

```
scores = cross_val_score(  
    pipeline,  
    X, y,  
    cv=kfold  
)
```

```
print(f"Mean accuracy = {np.mean(scores)}")
```

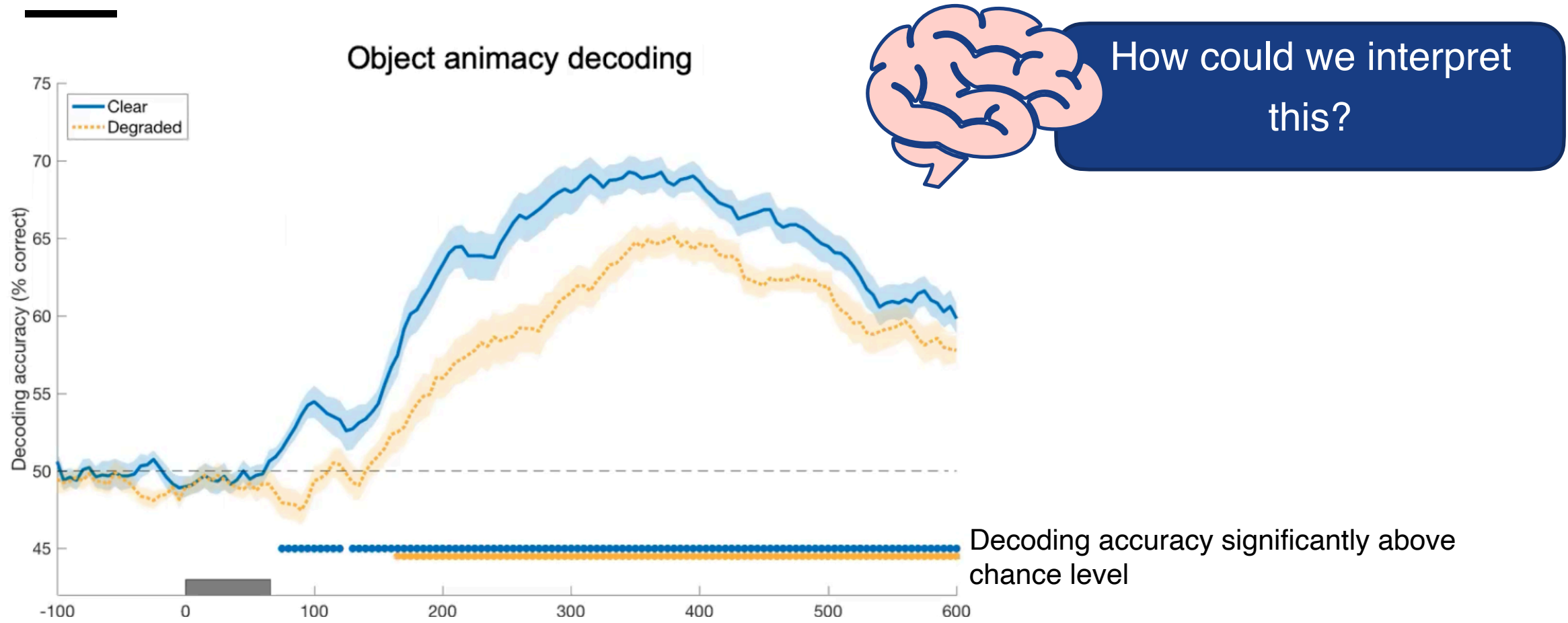


.... EXTENDING ACROSS TIME POINTS



```
scores = np.empty((ncv, X.shape[-1]))  
  
for t in range(X.shape[-1]):  
    X_t = X[:, :, t]  
    scores_t = cross_val_score(  
        pipeline,  
        X_t, y,  
        cv=kfold  
    )  
    scores[:, t] = scores_t
```

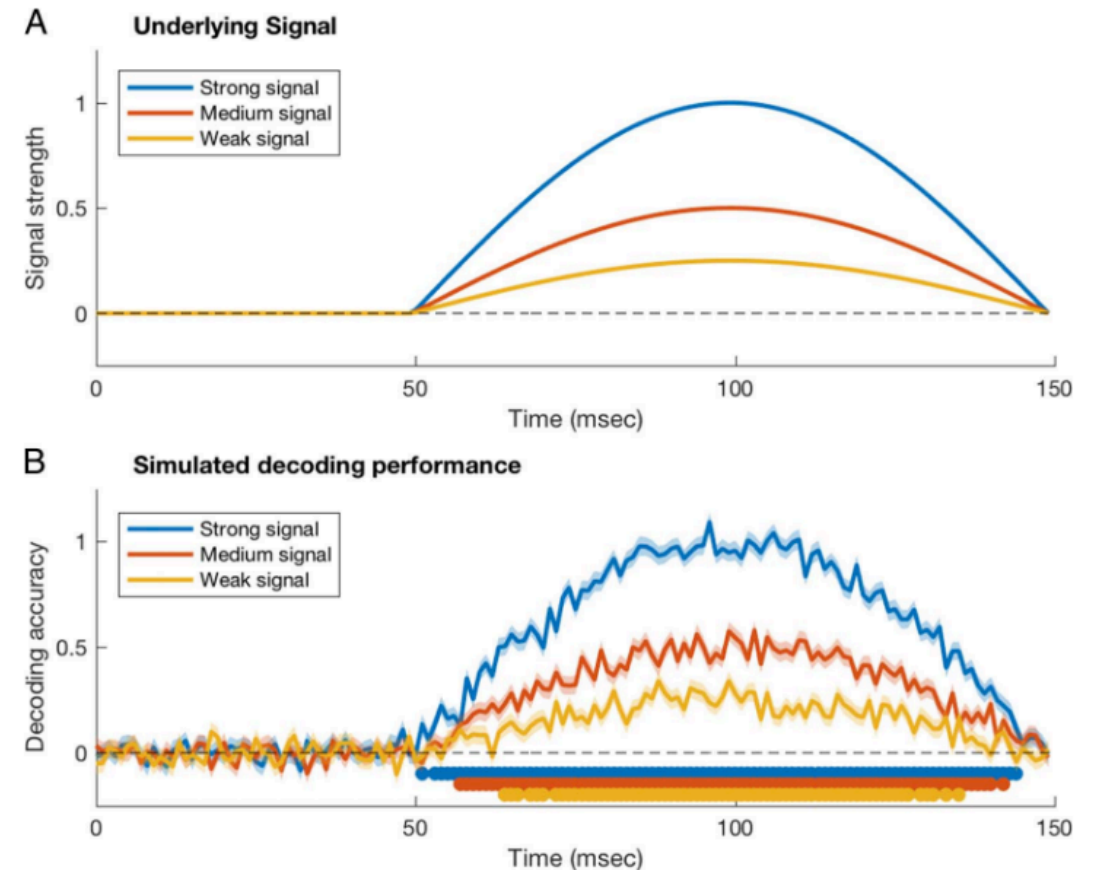
EXAMPLE: INTERPRETING ONSETS



EXAMPLE: INTERPRETING ONSETS

Simulated data with **same onset and peak latencies**, but **different peak strengths**

- Stronger signal resulted in earlier decoding onsets



Grootswagers et al. (2017). Decoding Dynamic Brain Patterns from Evoked Responses: A Tutorial on Multivariate Pattern Analysis Applied to Time Series Neuroimaging Data.



INFERENCE

- Repeat decoding analysis over subjects
- Test if group mean accuracy is above chance level
 - T-test
 - Permutation test
 - ...
- Correction for multiple comparisons across time points
 - Bonferroni
 - False discovery Rate (FDR)
 - ...



TEMPORAL GENERALISATION

Does the classifier trained at one time point generalise to other time points?

```
accuracies = np.zeros((ncv, X.shape[2], X.shape[2]))

# loop over the splits
for i, (train_inds, test_inds) in enumerate(kfold.split(X, y)):
    X_train = X[train_inds]
    X_test = X[test_inds]

    y_train = y[train_inds]
    y_test = y[test_inds]

    for t in range(X_train.shape[2]):
        X_train_scaled = scaler.fit_transform(X_train[:, :, t])

        # fit the classifier
        clf.fit(X_train_scaled, y_train)

        # loop over the timesamples for testing
        for tt in range(X_test.shape[2]):
            X_test_scaled = scaler.transform(X_test[:, :, tt])

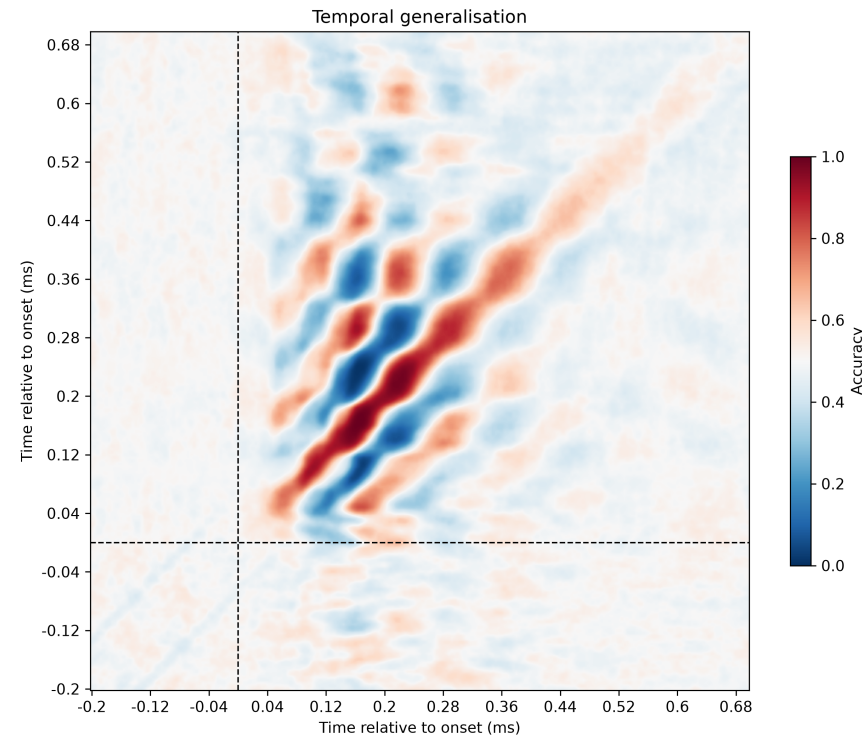
            accuracy = clf.score(X_test_scaled, y_test)
            accuracies[i, t, tt] = accuracy
```



TEMPORAL GENERALISATION



Example from our experiment (single subject), discriminating between mask and gabor patch. Why might we see this pattern?



EXTRA TRICKS AND TIPS

- Not covered in the notebooks but can help with decoding accuracy!
 - Sliding time window
 - Super trials



COMPARING SENSOR AND SOURCE SPACE DECODING

Sensor space

- Higher accuracy
- Lower spatial specificity

Source space

- Lower accuracy
- Higher spatial specificity
- Cross-participant generalisation by morphing to average brain

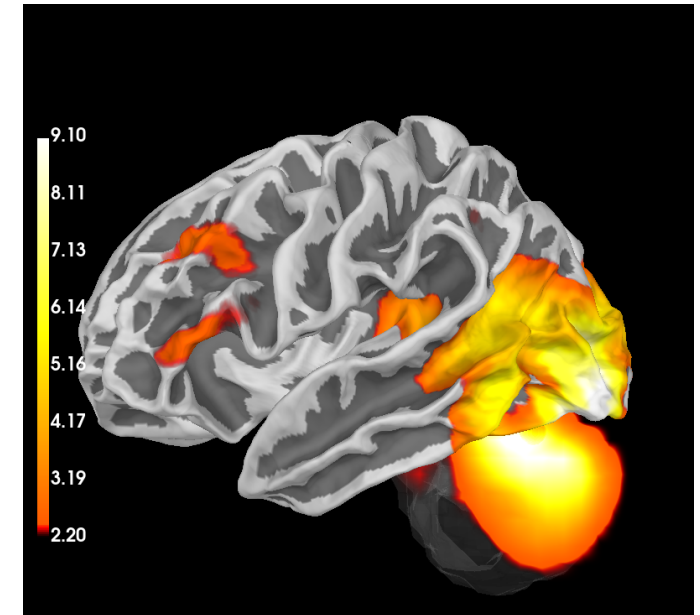
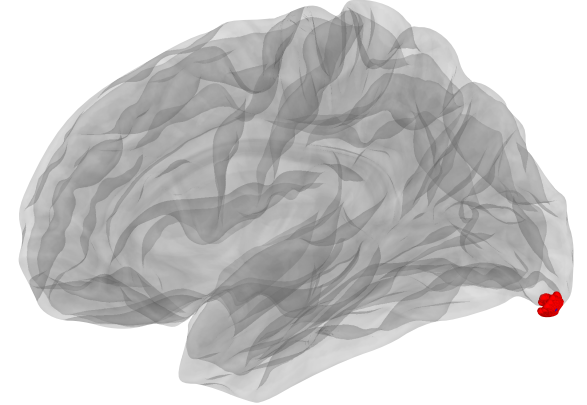


SOURCE SPACE

Simulated **occipital** source

- Source reconstruction using volumetric + surface source space show signal leakage
- Lead fields are correlated for nearby sources

Kruse et al., (in prep)



PERSPECTIVES

- Researcher-as-receiver vs cortex-as-receiver
 - Does the fact that we can decode between two conditions mean that the brain is using this information?

de-Wit et al., (2016). Is neuroimaging measuring information in the brain?



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